

AGENT INCIDENT REPORT

INC-2026-0247 | Generated 2026-02-10T03:17:42Z | Classification: HUMAN CONFIRMED

■ INCIDENT

This event resulted in confirmed economic loss. For near-misses blocked by guardrails, reports are flagged NEAR-MISS.

FAULT CLASS SPEC AMBIGUITY	IMPACT SCORE 4 / 5 HIGH	CONFIDENCE 0.84 HIGH	ESTIMATED LOSS \$48,868
LIABILITY POSTURE	PRIMARY RESPONSIBILITY Spec owner (refund_policy_v3)	CONTRIBUTING FACTOR Agent selection error (wrong transaction targeted)	POLICY OWNER payments-team@cme.com
AGENT refund-processor-v3	FRAMEWORK LangChain + OpenAI Tools	ENVIRONMENT production	TRIGGERED BY Customer support ticket #CS-88412

1. Executive Summary

On February 9, 2026 at 22:41 UTC, the **refund-processor-v3** agent autonomously issued a \$47,218.00 refund to customer account ending in 4491 via Stripe. The refund was triggered by customer support ticket #CS-88412, in which the customer reported being charged twice for an annual enterprise subscription.

The agent correctly identified the duplicate charge but applied the refund to the **original transaction** (\$47,218) rather than the **duplicate transaction** (\$47,218), effectively refunding both charges. The customer received their full subscription for free. The constraint system evaluated the refund amount against policy and passed, because the individual refund amount (\$47,218) was within the per-transaction limit (\$50,000). However, the policy contained no check for **cumulative refund exposure per customer per billing cycle**.

This is classified as **SPEC_AMBIGUITY**: the agent operated within its stated constraints, but the constraints failed to capture the intended business rule. The agent did not malfunction. The specification was incomplete.

2. Impact Assessment

Category	Amount	Status	Notes
Direct cash loss	\$47,218.00	Confirmed	Refund issued to original transaction
Remediation labor	~\$1,200	Estimated	3 engineers x 2hrs + customer outreach
Stripe reversal fee	\$450	Confirmed	Dispute + processing fees
Customer relationship	Unknown	Monitoring	Enterprise account, renewal in 4 months

Total estimated loss	\$48,868
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3. Timeline

22:41:03	Agent receives ticket #CS-88412 from support queue
22:41:04	Agent queries Stripe API: retrieves 2 charges for customer acct_4491
22:41:05	Agent identifies duplicate: charge ch_9xK (original, Feb 1) and ch_A2m (duplicate, Feb 3)
22:41:06	Agent evaluates refund_policy_v3: amount=\$47,218 < max_per_txn=\$50,000 → PASS
22:41:06	No cumulative exposure check exists in policy — gap not detected
22:41:07	Agent calls stripe.refund on ch_9xK (original) instead of ch_A2m (duplicate)
22:41:07	Stripe returns HTTP 200, refund re_Xp7 confirmed
22:41:08	Agent sends confirmation email to customer
22:47:00	Finance alert: cumulative refund for acct_4491 exceeds \$50k threshold
23:12:00	On-call engineer begins investigation
23:34:00	Root cause identified, remediation initiated

4. Root Cause Analysis

The root cause is a specification gap in **refund_policy_v3**. The policy defines a per-transaction refund limit (\$50,000) but contains no constraint on cumulative refund exposure per customer per billing cycle. The agent correctly evaluated the policy and received a PASS result, because the individual refund amount was within bounds.

A secondary contributing factor is that the agent selected the wrong transaction to refund. The customer reported a duplicate charge, and the correct action was to refund the **duplicate** (ch_A2m, Feb 3). Instead, the agent refunded the **original** (ch_9xK, Feb 1). This selection error, combined with the missing cumulative check, resulted in the customer receiving both the original service and a full refund of the original payment.

5. Fault Classification

Field	Value
fault_class	SPEC_AMBIGUITY
fault_subclass	refund_policy_boundary / cumulative_exposure
confidence	0.84
classifier	hybrid (rules + LLM, human confirmed)

status	HUMAN CONFIRMED
confirmed_by	maria.chen@acme.com
confirmed_at	2026-02-10T01:15:00Z

Classification Evidence

- Constraint refund_policy_v3 evaluated and returned PASS (margin to limit: \$2,782)
- Policy contains no cumulative_refund_per_customer_per_cycle constraint
- Agent action was consistent with policy as written
- Transaction selection error (original vs. duplicate) is secondary to policy gap

Counterfactuals

- If refund_policy_v3 included a cumulative exposure cap, the refund would have been blocked → AGENT_ERROR (downgraded)
- If the agent had selected ch_A2m instead of ch_9xK, the customer would have been correctly refunded → no incident
- If Stripe had returned an error, this would classify as TOOL_FAILURE

6. Contributing Factors

- **Policy gap:** refund_policy_v3 has no cumulative exposure limit. This is the primary cause.
- **Transaction disambiguation:** Agent prompt does not explicitly instruct preference for refunding the most recent (duplicate) charge when multiple charges exist.
- **No human-in-the-loop for high-value refunds:** The \$50,000 per-transaction threshold was set assuming single-refund scenarios. No escalation trigger for cumulative exposure.
- **Alerting delay:** Finance threshold alert fired 6 minutes after the refund. Earlier detection would have enabled reversal within Stripe's cancellation window.

7. Immediate Remediation

Action	Status	Owner	Completed
Initiated Stripe dispute for refund re_Xp7	Done	jake.r@acme.com	Feb 10, 00:12
Contacted customer to explain billing correction	Done	support@acme.com	Feb 10, 09:00
Suspended refund-processor-v3 from production	Done	maria.chen@acme.com	Feb 9, 23:40
Manual review of all refunds issued in last 72 hours	In progress	ops-team@acme.com	ETA Feb 11

8. Preventive Actions

- **Add cumulative exposure constraint** to refund_policy_v4: max cumulative refund per customer per 30-day cycle = \$50,000. Any refund that would exceed this triggers human escalation.
- **Add transaction selection rule:** when refunding a duplicate charge, always target the most recent transaction unless explicitly overridden.
- **Lower the autonomous refund threshold** from \$50,000 to \$10,000. Refunds above \$10,000 require human approval.
- **Add real-time cumulative exposure alerting** with a 60-second SLA (down from 6 minutes).

9. Policy & Constraint Changes

Diff: refund_policy_v3 → refund_policy_v4

```
"max_per_transaction": 50000,  
+ "max_per_transaction": 10000,  
+ "max_cumulative_per_customer_30d": 50000,  
+ "escalation_threshold": 10000,  
+ "require_human_approval_above": 10000,  
+ "duplicate_refund_preference": "most_recent_first",
```

Policy update deployed to staging: 2026-02-10T04:30:00Z. Production deployment pending verification (see Section 10).

10. Verification Plan

Test	Method	Status
Replay incident trace against refund_policy_v4	Deterministic replay	PASS ✓
Cumulative exposure exceeds \$50k → blocked	Synthetic test	PASS ✓

Single refund \$10,001 → escalates to human	Synthetic test	PASS ✓
Duplicate charge → most recent refunded first	Synthetic test	PASS ✓
Regression suite (412 existing test cases)	Automated	PASS ✓
72-hour canary deployment with shadow mode	Production canary	Scheduled Feb 11

11. Response Metrics

Time to detect	5 min 57 sec
Time to contain (agent suspended)	52 min 57 sec
Time to root cause	2 hr 11 min
Time to remediate (policy patch deployed to staging)	5 hr 49 min
Time to verify (regression suite passed)	6 hr 22 min

12. Appendix: Trace Excerpt

Canonical deterministic replay format (AgentIncident v0.1). Full trace available in incident store.

```
{
  "schema": "agentincident/v0.1",
  "incident_id": "INC-2026-0247",
  "incident_type": "INCIDENT",
  "replay_format": "deterministic_sequential",
  "trace": [
    {
      "seq": 1,
      "tool": "stripe.charges.list",
      "input": {"customer": "cus_4491", "limit": 10},
      "output": {"charges": ["ch_9xK ($47,218, Feb 1)", "ch_A2m ($47,218, Feb 3)"]},
      "ts": "2026-02-09T22:41:04.112Z",
      "duration_ms": 892,
      "hash": "sha256:a3f8c1d..."
    },
    {
      "seq": 2,
      "tool": "policy.evaluate",
      "input": {"policy": "refund_policy_v3", "amount": 47218},
      "output": {"pass": true, "margin": 2782, "gaps": ["no cumulative_exposure"]},
      "ts": "2026-02-09T22:41:06.004Z",
      "duration_ms": 41,
      "hash": "sha256:7b2e9d4..."
    }
  ]
}
```

```
    },
    {
      "seq": 3,
      "tool": "stripe.refund",
      "input": {"charge": "ch_9xK", "amount": 47218},
      "output": {"refund_id": "re_Xp7", "status": "succeeded"},
      "ts": "2026-02-09T22:41:07.331Z",
      "duration_ms": 1204,
      "hash": "sha256:e1d4f60...",
      "irreversible": true
    }
  ],
  "constraints": [
    {
      "policy": "refund_policy_v3",
      "version": "3.1.0",
      "eval_result": "pass",
      "margin_to_limit": 2782,
      "breach": false,
      "gaps_identified": ["no cumulative_exposure constraint"]
    }
  ],
  "classification": {
    "fault_class": "SPEC_AMBIGUITY",
    "fault_subclass": "refund_policy_boundary/cumulative_exposure",
    "fault_vector": "SPEC:refund_policy_v3/BOUNDARY:cumulative/ACTION:stripe.refund",
    "confidence": 0.84,
    "classifier": "hybrid",
    "status": "human_confirmed"
  },
  "liability": {
    "primary": "spec_owner:payments-team@acme.com",
    "contributing": ["agent:selection_error"]
  },
  "impact": {
    "score": 4,
    "loss_estimate_usd": 48868,
    "loss_breakdown": {
      "direct_cash": 47218,
      "remediation_labor": 1200,
      "stripe_fees": 450
    }
  },
  "response": {
    "time_to_detect_sec": 357,
    "time_to_contain_sec": 3177,
    "time_to_remediate_sec": 20940,
    "constraint_update": "refund_policy_v3 -> refund_policy_v4"
```

}
}

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